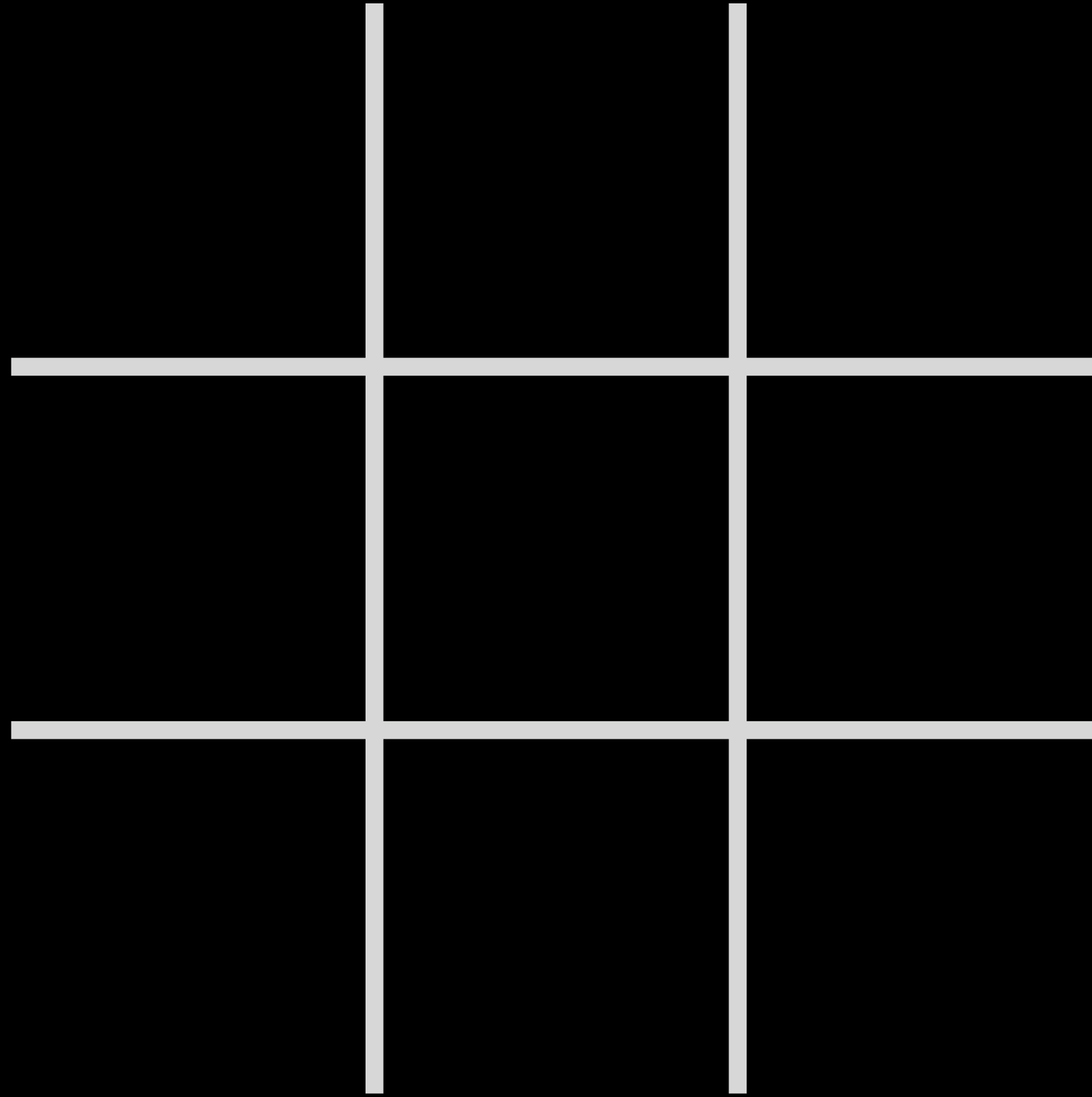
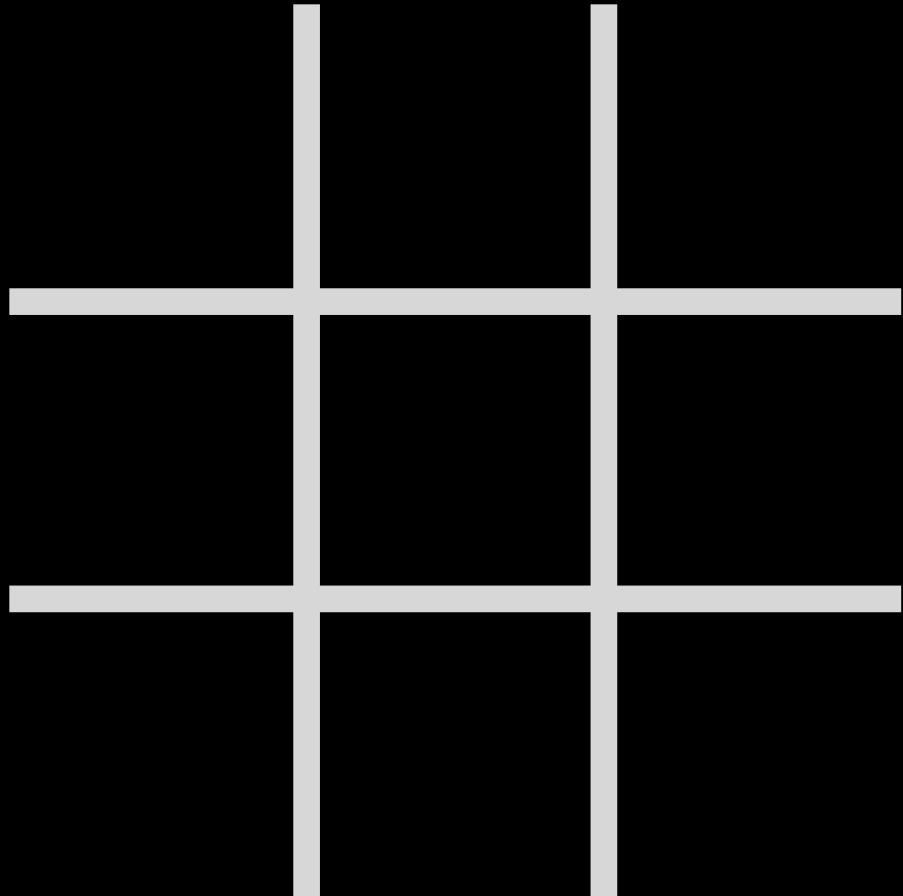
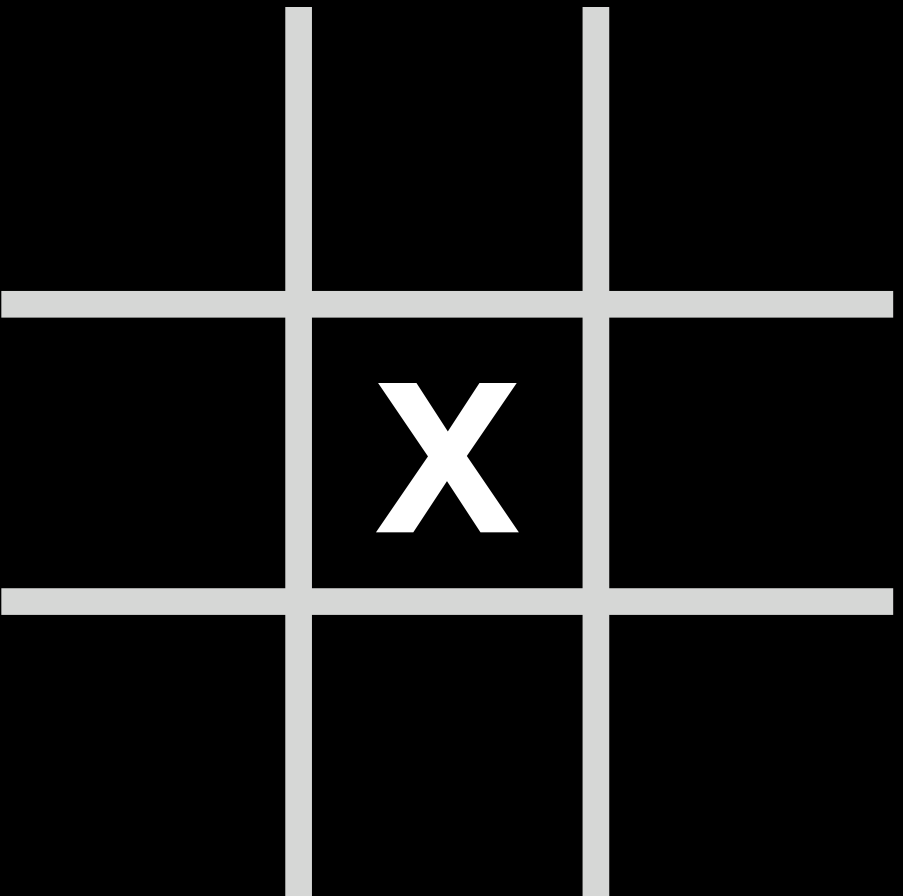


Initial State



PLAYER(*s*)

PLAYER() = **X**

PLAYER() = **O**

ACTIONS(s)

$$\text{ACTIONS}\left(\begin{array}{c|c|c} & \text{X} & \text{O} \\ \hline \text{O} & \text{X} & \text{X} \\ \hline \text{X} & & \text{O} \end{array}\right) = \left\{ \begin{array}{c|c|c} \text{O} & & \\ \hline & & \\ \hline & & \end{array}, \begin{array}{c|c|c} & & \\ \hline & & \\ \hline & \text{O} & \end{array} \right\}$$

RESULT(s, a)

$$\text{RESULT}\left(\begin{array}{c|c|c} & \text{X} & \text{O} \\ \hline \text{O} & \text{X} & \text{X} \\ \hline \text{X} & & \text{O} \end{array}, \begin{array}{c|c|c} \text{O} & & \\ \hline & & \\ \hline & & \end{array}\right) = \begin{array}{c|c|c} \text{O} & \text{X} & \text{O} \\ \hline \text{O} & \text{X} & \text{X} \\ \hline \text{X} & & \text{O} \end{array}$$

TERMINAL(s)

TERMINAL(

o		
o	x	
x	o	x

) = false

TERMINAL(

o		x
o	x	
x	o	x

) = true

UTILITY(s)

$$\text{UTILITY}\left(\begin{array}{c|c|c} \text{o} & & \text{x} \\ \hline \text{o} & \text{x} & \\ \hline \text{x} & \text{o} & \text{x} \end{array}\right) = 1$$

$$\text{UTILITY}\left(\begin{array}{c|c|c} \text{o} & \text{x} & \text{x} \\ \hline \text{x} & \text{o} & \\ \hline \text{o} & \text{x} & \text{o} \end{array}\right) = -1$$

O	X	O
O	X	X
X	X	O

VALUE: 1

PLAYER(s) = O

MIN-VALUE:
 $\emptyset$

	X	O
O	X	X
X		O

MAX-VALUE:
1

O	X	O
O	X	X
X		O

MAX-VALUE:
 $\emptyset$

	X	O
O	X	X
X	O	O

VALUE:
1

O	X	O
O	X	X
X	X	O

VALUE:
 $\emptyset$

X	X	O
O	X	X
X	O	O

PLAYER(s) = O

MIN-VALUE:
 $\emptyset$

	X	O
O	X	X
X		O

MAX-VALUE:
1

O	X	O
O	X	X
X		O

MAX-VALUE:
 $\emptyset$

	X	O
O	X	X
X	O	O

VALUE:
1

O	X	O
O	X	X
X	X	O

VALUE:
 $\emptyset$

X	X	O
O	X	X
X	O	O

PLAYER(s) = X

MAX-VALUE:
1

	X	O
O	X	
X		O

MIN-VALUE:
0

	X	O
O	X	X
X		O

MAX-VALUE:
1

O	X	O
O	X	X
X		O

VALUE:
1

O	X	O
O	X	X
X	X	O

MAX-VALUE:
0

	X	O
O	X	X
X	O	O

VALUE:
0

X	X	O
O	X	X
X	O	O

MIN-VALUE:
-1

X	X	O
O	X	
X		O

VALUE:
-1

X	X	O
O	X	O
X		O

MAX-VALUE:
0

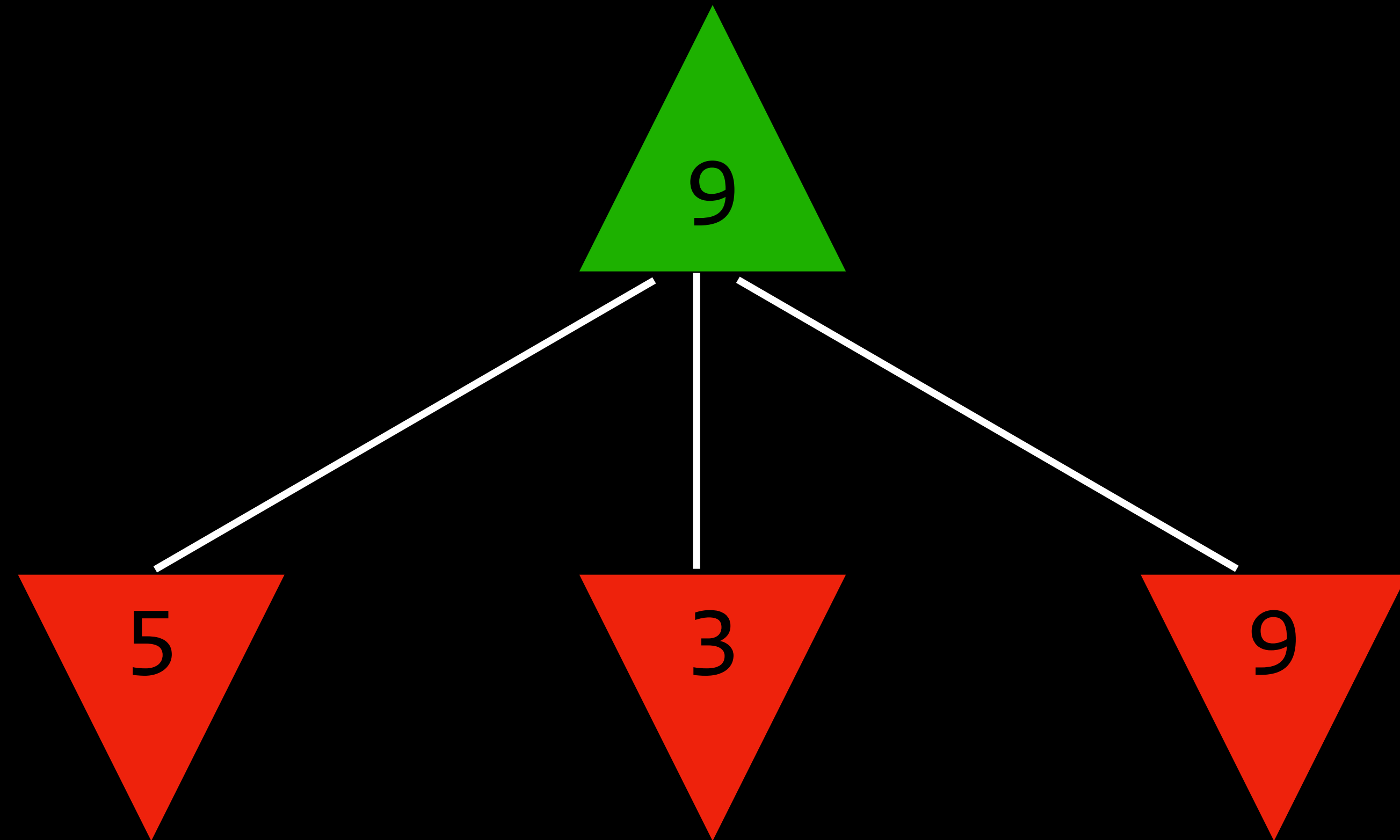
X	X	O
O	X	
X	O	O

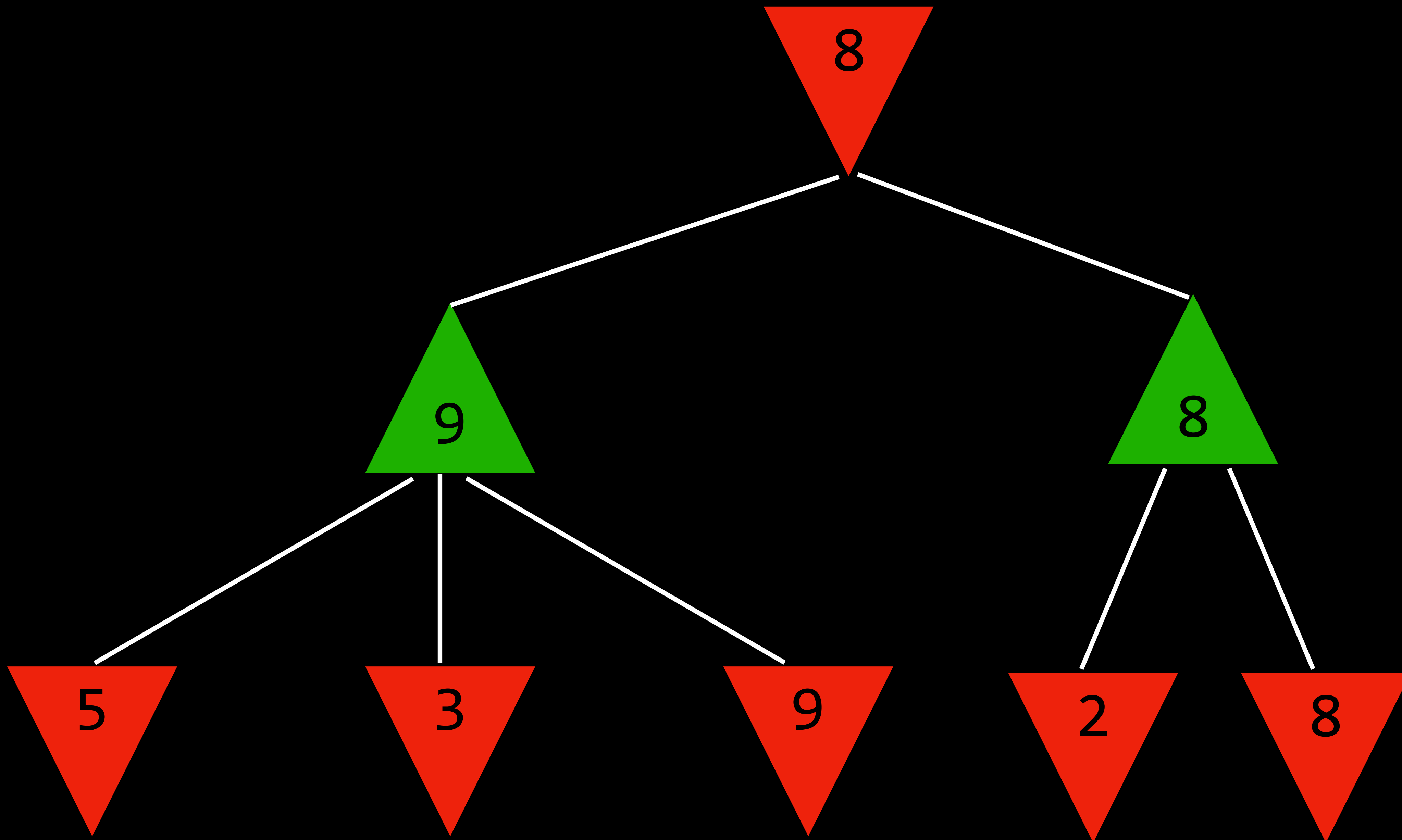
VALUE:
0

X	X	O
O	X	X
X	O	O

VALUE:
1

	X	O
O	X	
X	X	O





Minimax

- Given a state s :
 - **MAX** picks action a in $\text{ACTIONS}(s)$ that produces highest value of $\text{MIN-VALUE}(\text{RESULT}(s, a))$
 - **MIN** picks action a in $\text{ACTIONS}(s)$ that produces smallest value of $\text{MAX-VALUE}(\text{RESULT}(s, a))$

Minimax

```
function MAX-VALUE(state):  
  if TERMINAL(state):  
    return UTILITY(state)  
   $v = -\infty$   
  for action in ACTIONS(state):  
     $v = \text{MAX}(v, \text{MIN-VALUE}(\text{RESULT}(\textit{state}, \textit{action})))$   
  return  $v$ 
```


Minimax

```
function MIN-VALUE(state):  
  if TERMINAL(state):  
    return UTILITY(state)  
   $v = \infty$   
  for action in ACTIONS(state):  
     $v = \text{MIN}(v, \text{MAX-VALUE}(\text{RESULT}(\textit{state}, \textit{action})))$   
  return  $v$ 
```